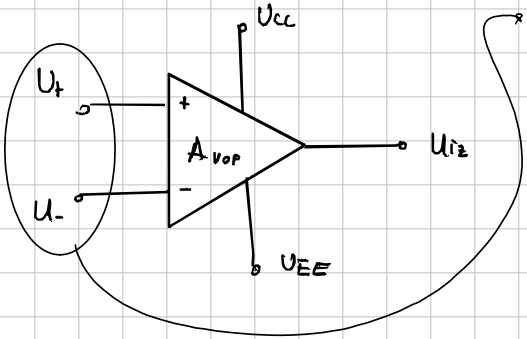
## El. simbol




$$U_{iz} = A_{vop}\,(\overset{\oplus}{U_+} - \overset{\ominus}{U_-}) = A_{vop}\cdot U_d$$

$U_- = \phi \rightarrow U_{iz} = A_{vop}\cdot U_+ \xrightarrow{\oplus}$ u fazi

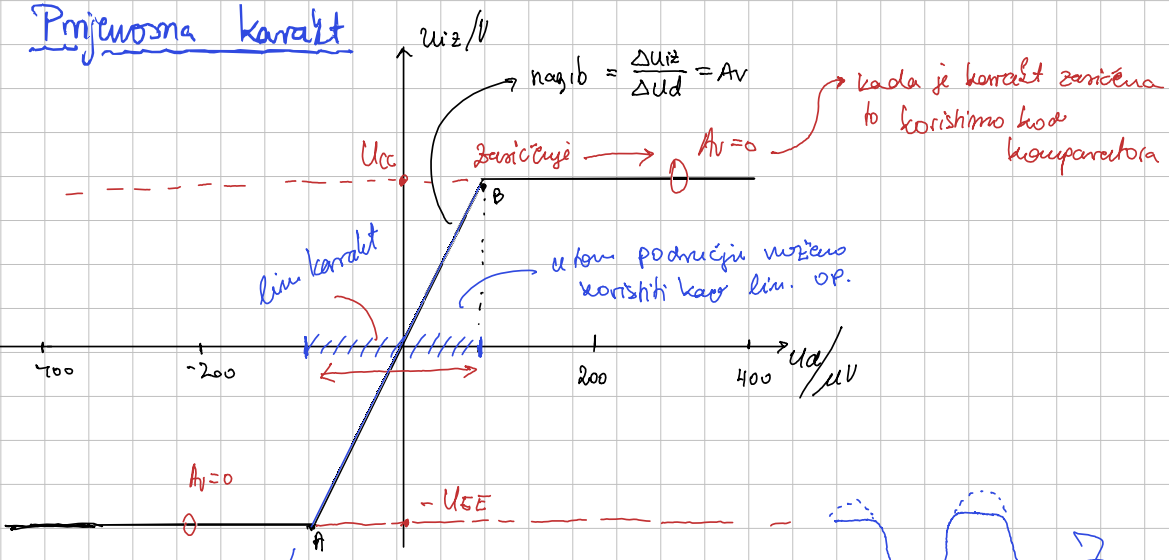
$U_+ = \phi \rightarrow U_{iz} = -A_{vop}\cdot U_- \xrightarrow{\ominus}$ u protufazi

| Svojstva | Idealno pojačalo | Realno pojačalo |
|---|---|---|
| Naponsko poj. | $\infty$ | $10^4 - 10^6$ (80 ~ 120 dB) |
| $R_{ul}$ | $\infty$ | $1M\Omega$ i više |
| $R_{iz}$ | 0 | $100\,\Omega$ i manje |
| $f_g$ | $\infty$ | 10 Hz i više |

idealno naponsko

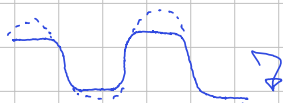
↳ posljedica frekv. kompenzacije

## Prijenosna karakt

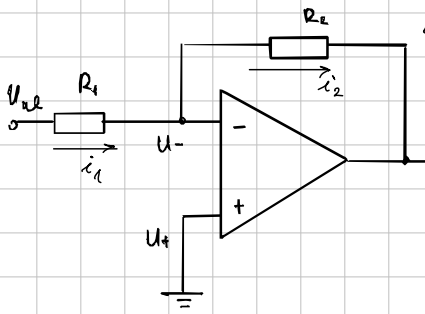



kada je karakt zasićena to koristimo kao komparatora

prelaskom preko točke A i B dolazi do „rezanja" sinusnog signala → liči na pravokutni

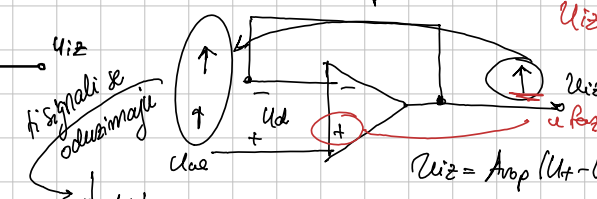


# INVERTIRAJUĆE POJAČALO




→ izlaz spojen s ulazom = povratna veza

Zašto bitno?




$U_{iz}$ u fazi sa $U_{ul}$

$U_{iz}$ u fazi sa

$U_{iz} = A_{op}(U_+ - U_-)$

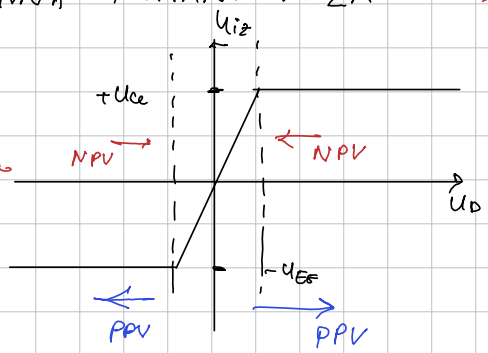
ti signali se oduzimaju

↓ $U_d$ se smanjuje

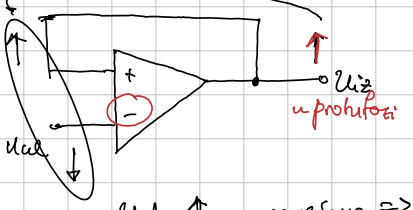
⇒ NEGATIVNA POVRATNA VEZA (NPV)

→ takvo djelovanje nas tjera da se radna točka ostavi u lin. područje karakt

↳ NPV osiguravaju ostajanje u lin. području (lin. OP)




II. način spajanja

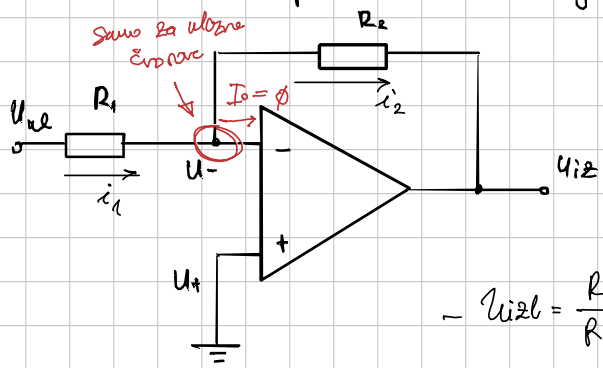



↗ tjera u zasićenje

$U_d$ ↑ se povećava ⇒ POZITIVNA POVRATNA VEZA (PPV)

$U_{iz} = A_{op}(U_+ - U_-) \longrightarrow U_+ - U_- = \frac{U_{iz}}{A_{op}} \xrightarrow{A \to \infty} U_+ = U_-$ negdje oko ishodišta

Kada analiziramo pretpostavimo da je to pojačalo s $R_{ul}$ idealno → $R_{ul} = \infty$




$I_0 = \emptyset \longrightarrow$ Struja ne ide tamo gdje je veliki otpor

↳ $I_1 = I_2$

$$\frac{U_{ul} - U_-}{R_1} = \frac{U_- - U_{izl}}{R_2}$$

$$-U_{izl} = \frac{R_2}{R_1}(U_{ul} - U_-) - U_-$$

$$U_{izl} = \frac{-R_2}{R_1}(U_{ul} - U_-) + U_- = U_- \cdot \left(1 + \frac{R_2}{R_1}\right) - \frac{R_2}{R_1} \cdot U_{ul} \quad (1)$$

$$U_{iz} = A_{op}(U_+ - U_-) = \{U_+ = 0\} = -A_{op} \cdot U_- \Rightarrow U_- = -\frac{U_{iz}}{A_{op}} \quad (2)$$

$$U_{iz} = \frac{-1}{A_{op}}\left(1 + \frac{R_2}{R_1}\right) \cdot U_{iz} - \frac{R_2}{R_1} \cdot U_{ul}$$

Korekcijski faktor

$$U_{iz}\left[1 + \frac{1}{A_{op}}\left(1 + \frac{R_2}{R_1}\right)\right] = \frac{-R_2}{R_1} \cdot U_{ul} \Rightarrow \boxed{A_v = \frac{U_{iz}}{U_{ul}} = -\frac{R_2}{R_1} \cdot \frac{1}{1 + \frac{1}{A_{op}}\left(1 + \frac{R_2}{R_1}\right)}}$$

$\times$ $A_{op} \longrightarrow \infty \longrightarrow A_v = \frac{-R_2}{R_1}$ idealno pojačanje

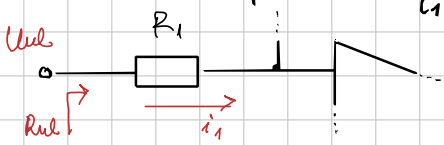
$|A_v| \ll A_{op}$ → posljedica neg. povr. veze (NPV)

zbog velikog pojačanja $A_{op} \rightarrow U_+ = U_-$

↳ poviđeni K.S.

→ posljedica velikog $A_{op}$ !! ne $R_{ul}$ (usmeni)

Ulazni otpor: $R_{ul} = \frac{U_{ul}}{i_1}$




$$R_{ul} = \frac{i_1 \cdot R_1 + \overset{=0}{U_-}}{i_1} = \frac{i_1 \cdot R_1}{i_1}$$

$$\Rightarrow \boxed{R_{ul} = R_1}$$

Pojednostavljena analiza sklopova s OP:

- $R_{ul} = \infty \longrightarrow I_0 = \emptyset$
- $A_{vop} = \infty \longrightarrow U_+ = U_-$

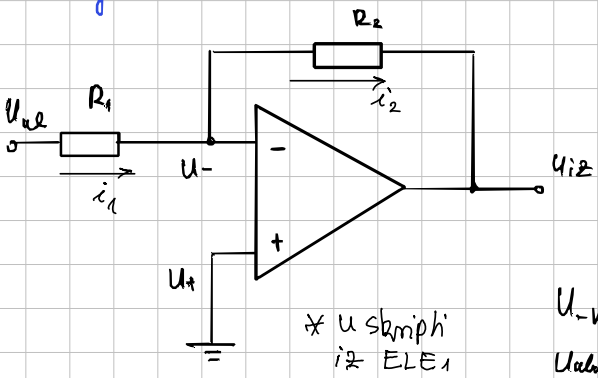
$\Rightarrow$ <u>pretpostavka IDEALNO OP:</u> $R_{ul} = \infty$, $A_{vop} = \infty \Rightarrow i_1 = i_2$

$$\frac{U_{ul} - U_-}{R_1} = \frac{U_- - U_{iz}}{R_2} \Rightarrow \frac{U_{ul} - \emptyset}{R_1} = \frac{\emptyset - U_{iz}}{R_2}$$

$U_+ = U_- = \emptyset$ (posljedica $A_{vop}$) $\Rightarrow$ $\boxed{\frac{U_{iz}}{U_{ul}} = \frac{-R_2}{R_1}}$

$\rightarrow$ budući da je idealizirano, mora imati neku pogrešku

Primjer 10.1)




$R_1 = 1\,k\Omega$ $R_2 = 100\,k\Omega$

$A_v = ?$, $A_{vop} = 10^3, 10^4, 10^5$

+ za svako odrediti pogrešku $A_v$ u odnosu na $A_{v\,idealno} = \frac{-R_2}{R_1}$

$U_{-m} = ?$ invertirajućeg ulaza op

$U_{ulm} = 0.1\,V$ (sinusni napon) ?

* u skripti iz ELE1

$$A_{vi} = \frac{-R_2}{R_1} = \frac{-100}{1} = -100$$

$$A_v = \frac{U_{iz}}{U_{ul}} = \boxed{\frac{-R_2}{R_1}} \cdot \frac{1}{1 + \frac{1}{A_{vop}}\left(1 + \frac{R_2}{R_1}\right)}$$

$x$

$$\varepsilon_r = \frac{A_v - A_{vi}}{A_{vi}} \text{ (rel. pogreška)}$$

*TUELE

$$U_{-m} = \frac{U_{izm}}{-A_{vop}} = \frac{A_v \cdot U_{ulm}}{-A_{vop}}$$

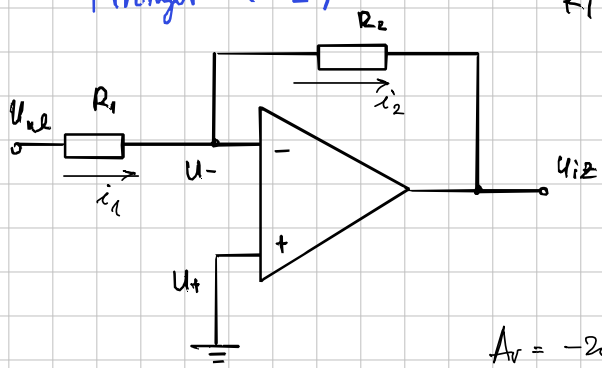
| $A_{vop}$ | $A_{vi}$ | $\varepsilon_r$ | $U_{-m}$ | |
|---|---|---|---|---|
| $10^3$ | | $-90{,}8$ | $-10.1\%$ | $9{,}08\,mV$ |
| $10^4$ | $-100$ | $-99{,}0$ | $-1{,}0\%$ | $0{,}99\,mV$ |
| $10^5$ | | $-99{,}9$ | $-0.1\%$ | $0{,}1\,mV$ |

$A_{vop}\uparrow$, pojačanje je preciznije

$A_{vop}\uparrow$; $x \rightarrow 1$

$\frac{R_2}{R_1}\downarrow \Rightarrow x \rightarrow 1$

## Primjer 10.2)




$R_1, R_2 = ?$

$R_{ul} = 2\,k\Omega$, $A_v = -200$

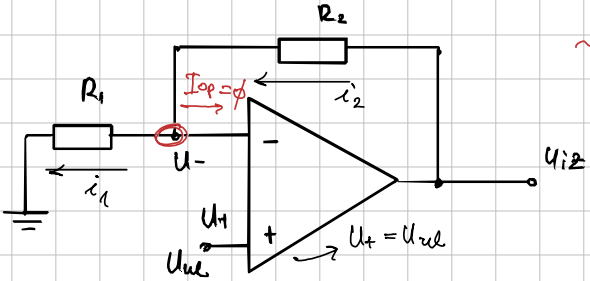
OP = idealno

$R_{ul} = 2\,k\Omega \Longrightarrow R_1 = 2\,k\Omega$

$$A_v = -200 = \frac{-R_2}{R_1} \Longrightarrow R_2 = 200 \cdot R_1$$

$$R_2 = 400\,k\Omega$$

## NEINVERTIRAJUĆE POJAČALO




$R_{ul} = \infty \Longrightarrow I_{op} = \phi$

$A_{vop}$ je konačno

$i_1 = i_2$

$$\frac{U_-}{R_1} = \frac{u_{iz} - U_-}{R_2} \quad (1)$$

$$u_{iz} = A_{vop}(U_+ - U_-) = A_{vop}(U_{ul} - U_-) \quad (2)$$

$$(2) \rightarrow U_- = U_{ul} - \frac{u_{iz}}{A_{vop}}$$

$$(1) \rightarrow u_{iz} = \frac{R_2}{R_1} \cdot U_- + U_- = U_-\left(1 + \frac{R_2}{R_1}\right)$$

$$u_{iz} = \left(U_{ul} - \frac{u_{iz}}{A_{vop}}\right) \cdot \left(1 + \frac{R_2}{R_1}\right)$$

$$U_{ul}\left(1 + \frac{R_2}{R_1}\right) = u_{iz}\left(1 + \frac{1}{A_{vop}}\left(1 + \frac{R_2}{R_1}\right)\right)$$

$$\Longrightarrow \frac{u_{iz}}{U_{ul}} = A_v = \boxed{\left(1 + \frac{R_2}{R_1}\right)} \cdot \frac{1}{1 + \frac{1}{A_{vop}}\left(1 + \frac{R_2}{R_1}\right)}$$

pojačanje idealnog op

korekcijski faktor

$A_{vop} \rightarrow \infty$

$$\hookrightarrow \boxed{A_v = A_{vi} = \left(1 + \frac{R_2}{R_1}\right)}$$

pojačanje je pozitivno

### Idealno OP

$R_{ul} = \infty \longrightarrow \frac{U_-}{R_1} = \frac{u_{iz} - U_-}{R_2}$

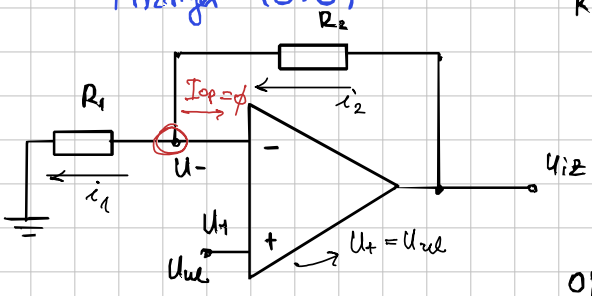
$A_{vop} = \infty \longrightarrow U_- = U_+ = U_{ul}$

$$\frac{U_{ul}}{R_1} = \frac{u_{iz} - U_{ul}}{R_2}$$

$$U_{ul}\left(1 + \frac{R_2}{R_1}\right) = u_{iz}$$

$$\Longrightarrow A_v = \frac{u_{iz}}{U_{ul}} = \left(1 + \frac{R_2}{R_1}\right)$$

## Primjer (10.3)




$R_1 = 3k\Omega$ $R_2 = 50k\Omega$

$A_v = ?$ $U_{-m} = ?$ $U_{izm} = ?$

$I_{1m} = ?$ $I_{2m} = ?$

$U_{ulm} = 0.1V$

OP je idealno

$$A_v = 1 + \frac{R_2}{R_1} = 1 + \frac{50}{3} = \underline{\underline{17,7}}$$ (>0 jer je neinvert...)

$$\frac{U_{izm}}{U_{ulm}} = A_v \Rightarrow U_{izm} = A_v \cdot U_{ulm} = 17,7 \cdot 0,1 \Rightarrow U_{izm} = 1,77V$$

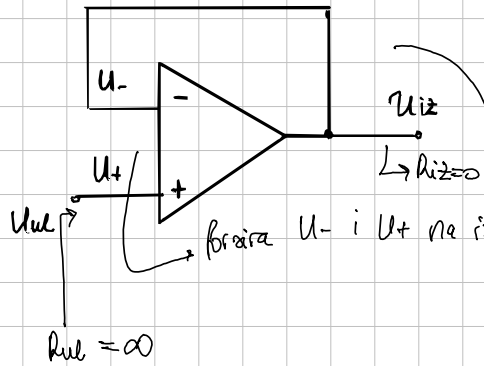
$$U_{-m} = U_{+m}$$ (jer $A_{vop} = \infty$) $\rightarrow U_{-m} = U_{+m} = U_{ulm} = 0.1V$

$$I_{1m} = \frac{U_{-m} - \emptyset}{R_1} = \frac{0,1}{3} = 0,033mA \rightarrow \boxed{I_{1m} = 33\mu A} \Rightarrow I_{2m}^{*}$$

$R_{ul\,op} = \infty$

## Naponsko sljedilo - specijalan slučaj neinvert pojačala

ut $R_2 = 0$ i $R_1 = \infty$



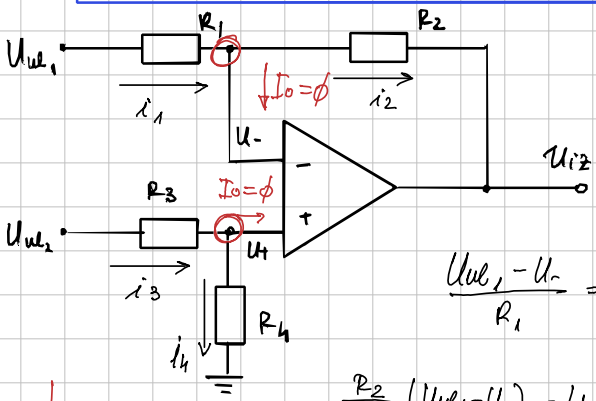

$$A_v = \frac{U_{iz}}{U_{ul}} = 1$$

pojačalo NPV → vodi nas u lin. područje

prisila U- i U+ na isti potencijal

=> odijelni stupanj ili transformator impedancija → slično kao SZC / SZD

# DIFERENCIJSKO POJAČALO




OP idealno: $R_{ul}=\infty$

$A_{vop}=\infty$

$i_1=i_2$

$$\frac{U_{ul_1}-U_-}{R_1}=\frac{U_--U_{iz}}{R_2} \quad \Big/ \cdot R_2$$

$$\frac{R_2}{R_1}(U_{ul_1}-U_-)=U_--U_{iz} \Rightarrow U_{iz}=U_--\frac{R_2}{R_1}(U_{ul}-U_-)$$

vrijedi superpozicija

$$\boxed{U_{iz}=U_-\left(1+\frac{R_2}{R_1}\right)-\frac{R_2}{R_1}U_{ul_1}} \quad (1)$$

$U_{iz}=$ ⟨pojačanje neinv. pojačala⟩ · ⟨ulaz dovedan na neinv. ulaz⟩ +

+ ⟨pojačanje inv. pojačala⟩ · ⟨napon na ulazu inv. pojačala⟩

$U_-$
$\|$

$$U_+=U_{ul_2}\cdot\frac{R_4}{R_3+R_4} \quad (2)$$

$(2)\rightarrow(1)$ $$U_{iz}=U_{ul_2}\cdot\frac{R_4}{R_3+R_4}\cdot\frac{R_1+R_2}{R_1}-\frac{R_2}{R_1}U_{ul}$$

Želimo dobiti da je $U_{iz}=f(U_{ul_2}-U_{ul_1})$

$$\frac{R_4}{R_3+R_4}\cdot\frac{R_1+R_2}{R_1}=\frac{R_2}{R_1}$$

mora: $R_2=R_4$

$R_3+R_4=R_1+R_2$

$R_3=R_1$

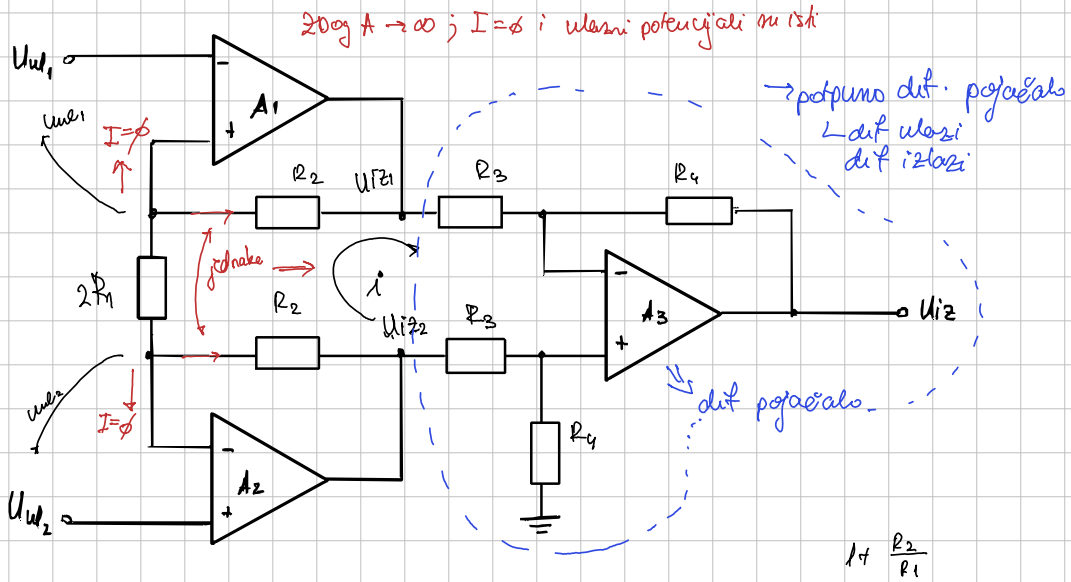
$$\Rightarrow \boxed{U_{iz}=\frac{R_2}{R_1}(U_{ul_2}-U_{ul_1})}$$

* ovo pojačalo ima konačne ulazne otpore → nije baš dobro

izbjegnemo ako koristimo instrumentacijsko pojačalo

# INSTRUMENTACIJSKO POJAČALO

Zbog $A \to \infty$; $I = \emptyset$ i ulazni potencijali su isti



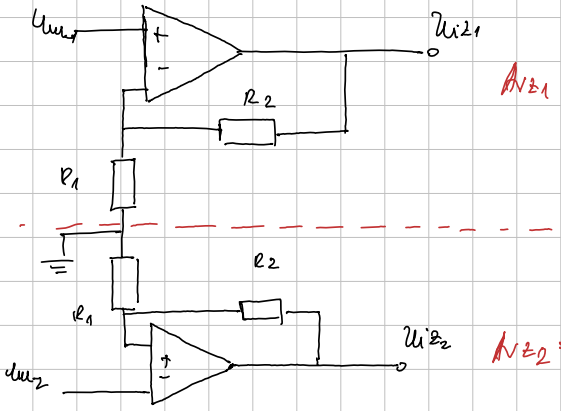

$$i = \frac{U_{iz_2} - U_{iz_1}}{2R_1 + 2R_2} = \frac{U_{ul_2} - U_{ul_1}}{2R_1} \Rightarrow (U_{iz_2} - U_{iz_1}) = \frac{R_1 + R_2}{R_1}(U_{ul_2} - U_{ul_1}) \quad (1)$$

$$1 + \frac{R_2}{R_1}$$

$$U_{iz} = \frac{R_4}{R_3}(U_{iz_2} - U_{iz_1}) \quad (2)$$

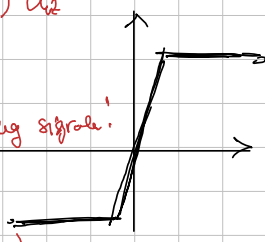
$$\boxed{U_{iz} = \frac{R_4}{R_3} \cdot \frac{R_1 + R_2}{R_1}(U_{ul_2} - U_{ul_1})}$$

prijenosna fja INSTRUM. POJ.




$$A_{z1} = \frac{U_{iz1}}{U_z} = \dots = \left(1 + \frac{R_2}{R_1}\right) U_z$$

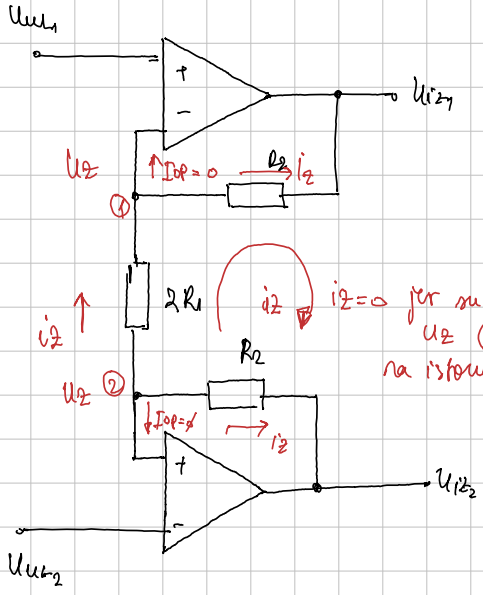
$A_z = A_{vd}$ → pojačanje zajedničkog signala!



$$A_{z2} = \frac{U_{iz_2}}{U_z} = \dots = \left(1 + \frac{R_2}{R_1}\right) U_z$$

$$U_z = \frac{U_{ul_1} + U_{ul_2}}{2}$$

(zajednički)




$i_z = \emptyset \Rightarrow U_{iz_1} = U_{iz_2} = U_z$

$A_{z1} = A_{z2} = \underline{\underline{1}}$

Primjer 10.4) $R_1 = 15k\Omega$ $R_2 = 150k\Omega$ $R_3 = 15k\Omega$ $R_4 = 30k\Omega$

*Neinvert OP $A_1$ i $A_2$ $\rightarrow U_{ul_1} = 2{,}5V$

$U_{ul_2} = 2{,}25V$

$U_{iz_1}, U_{iz_2}, U_{iz} = ?$

$U_{iz_2} = U_{iz_1} - 0.275V$ , $i = \frac{U_{ul_2} - U_{ul_1}}{2R_1} = -8{,}33\mu A$

$i = \frac{U_{ul_1} - U_{iz_1}}{R_2} \rightarrow U_{iz_1} = U_{ul_1} - i \cdot R_2 = \boxed{3.75V = U_{iz_1}}$

$\boxed{U_{iz_2} = U_{ul_2} + i \cdot R_2 = 1V}$

$U_{iz} = \frac{R_4}{R_3}(U_{iz_2} - U_{iz_1}) = \dots \Rightarrow \boxed{U_{iz} = -5.5V}$